MILITARY SPECIFICATION
CIRCUIT CARD ASSEMBLY: 11746286
POSITION-2

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the detail requirement for one type of circuit card assembly. The requirements of this specification together with the general requirements of specification MIL-C-48411 comprise the complete requirements for Position - 2.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-C-48411	Circuit Card Assemblies, General Specification for

DRAWINGS

11746286	Circuit Card Assembly, Position - 2.

(Copies of specifications, standards, drawings, and publications required by supplier in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Item definition.- This assembly is the second of four (4) single ended printed circuit boards consisting of operational amplifiers, gates, a digital to analog converter necessary to provide target azimuth position information.

3.1.1 Major component list.- The components of this assembly shall be as specified on the Engineering Parts List (EPL) 11746286.

3.2 Characteristics.

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3.2.1 Performance.- The performance requirements specified herein shall be met on every assembly at standard ambient conditions prior to Government acceptance.

3.2.1.1 Power Supply.- The requirements of 3.2 shall be met with the following power supply voltages:

a. +5 VDC
b. +15 VDC
c. -5 VDC
d. -10 VDC
e. -15 VDC

3.2.1.2 Integrating network.- The integrating network shall integrate inputs on Pins 30&20 and yield the output on Pin 28 as shown in Table 1.

3.2.1.3 D/A Converter.- The inputs to the D/A converter are provided by the synchronous 4 - bit up/down counters; the D/A converter output is Pin 22. The D/A converter test procedure is described in paragraph 4.2.2.3.2.

3.2.1.4 Azimuth zero network.- With the inputs applied as described in the azimuth test procedure, paragraph 4.2.2.3.3, and when the D/A converter output (Pin 22) exceeds +8.92 VDC, the azimuth indicator lamp (Pin 36) shall start and remian blinking until Pin 22 decreases to -8.92 VDC at which the lamp shall go out.

3.2.2 Environmental.- The assembly shall meet the environmental requirements of specification MIL-C-48411.

3.2.3 Physical Characteristics.- The physical characteristics shall be as specified on drawing 11746286 and the engineering parts list.

3.3. Design and Construction.

3.3.1 Production Drawings.- This assembly shall be fabricated in accordance with the requirements of specification MIL-C-48411, Drawing 11746286 and associated lists.

3.3.2 Standard of Manufacture.- The standard of manufacture shall be those specified in specification MIL-C-48411.

3.3.3 Workmanship.- Thw workmanship shall be as specified in specification MIL-C-48411.

3.3.4 Preproduction sample.- Preproduction sampling shall be as specified in specification MIL-C-48411.

4. QUALITY ASSURANCE PROVISIONS

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4.1 General.

4.1.1 Responsibility for inspection.- The responsibility for inspection shall be as specified in specification MIL-G-48411.

4.1.2 Special test and examination.- The assembly shall undergo first article and counter testing in accordance with 4.1.2 of specification MIL-G-48411.

4.2 Quality conformance inspection.- The assembly shall be inspected and tested in accordance with 4.2 of specification MIL-G-48411.

4.2.1 Test equipment.(see Figure 1)

4.2.1.1 Power Supply.- The power supplies shall be as follows:

a. +10.000 VDC ± 1mVDC, 100mA or greater and a ripple of less than 10 mVRMS.

b. +20.0VDC ±0.1VDC, 100mA or greater and a ripple of less than 10 mVRMS.

c. -10.000 VDC ± 1mVDC, 100 mA or greater and a ripple of less than 10 mVRMS.

d. -15.000 VDC ± 1mVDC, 100mA or greater and a ripple of less than 10 mVRMS.

e. -20.0VDC ± 0.1VDC, 100mA or greater and a ripple of less than 10 mVRMS.

4.2.1.2 Pulse Generator.- The generator shall be capable of producing a single 10 nanoscond pulse, a 15 KHz ± 1 Hz clock pulse, and a 60 KHz ±1Hz clock pulse; each shall have an amplitude of 0.0VDC to 5.0VDC ± 0.1VDC.

4.2.1.3 Oscilloscope.- Rise time of scope plus associated measuring probe of 50 nanoseconds or less, sweep period of 50 nanoseconds per centimeter or greater, input resistance greater than 8 megohms, input capacitance less than 25 picafarads.

4.2.1.4 Dummy Loads.- The dummy loads shall be three 10K ohm resistors; and one MS25-327 indicating lamp or its equivalent.

4.2.1.5 Digital Voltmeter.- The digital voltmeter shall have a voltage accuracy and resolution of 0.1 percent or better from zero to 100 VDC. Input inpedance of 10 megohms or greater on all voltage scales.

4.2.2 Performance.- Unless otherwise specified, the performance tests specified herein shall be satisfactorily met at standard ambient conditions. Standard ambient conditions for the purpose of these tests shall be defined as:

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a. Temperature between 60°F and 90°F
b. Barometric pressure between 28 and 32 inches of mercury.
c. Relative Humidity (RH) up to 90%.

4.2.2.1 Power Supply.- All requirements shall be met with the following power supplies:

a. +5.000VDC ±1mVDC
b. +15.0VDC ± 0.1 VDC
c. -5.000 VDC ± 1 mVDC
d. -10.000 VDC ± 1mVDC
e. -15.0VDC ± 0.1 VDC

4.2.2.2 Digital Output.- The "0" or false output level shall be between 0.00 and 0.40 VDC. The "1" or true output level shall be between +2.40 and +5.00 VDC.

4.2.2.3 Test Set-up.- The test of the circuit card assembly shall be performed with the inputs and outputs as diagrammed in Figure 1 unless otherwise specified.

4.2.2.3.1 Integrating network test.- The integrating network shall integrate input Pins 30 & 20, yielding the output on Pin 28 as shown in Table I.

TABLE I

Inputs (VDC)		Output (VDC)
Pin 30	Pin 20	Pin 28
-10V±1mV	0V	2.72 ± .1
0V	-10V±1mV	-2.72 ± .1
5V±mV	-5V ±1mV	0.00 ± .1

4.2.2.3.2 D/A Converter Test.- When the inputs are applied as described, the D/A converter output Pin 22 shall be as stated in the below test procedure.

D/A Converter Test Procedure

With a logic 1 applied to Pin 5, reset "0" at Pin 2, output Pin 22 shall be 0V ± 50 mV. Apply a 10 nanosecond Logic 1 pulse to Pin 1, output Pin 22 shall be 5mV ±mV less than the previous value. Apply several additional 10 nanosecond Logic 1 pulses to Pin 22, reading the output voltage on Pin 22 after each pulse. Each successive Pin 22 reading shall be 5 mV ± 1mV less than the previous reading.

Apply a 60 Hz ± 1Hz clock pulse to Pin 1. Output Pin 22 voltage shall decrease to -10V ± .25V then switch to +10V ± .25V then decrease towards 0V.

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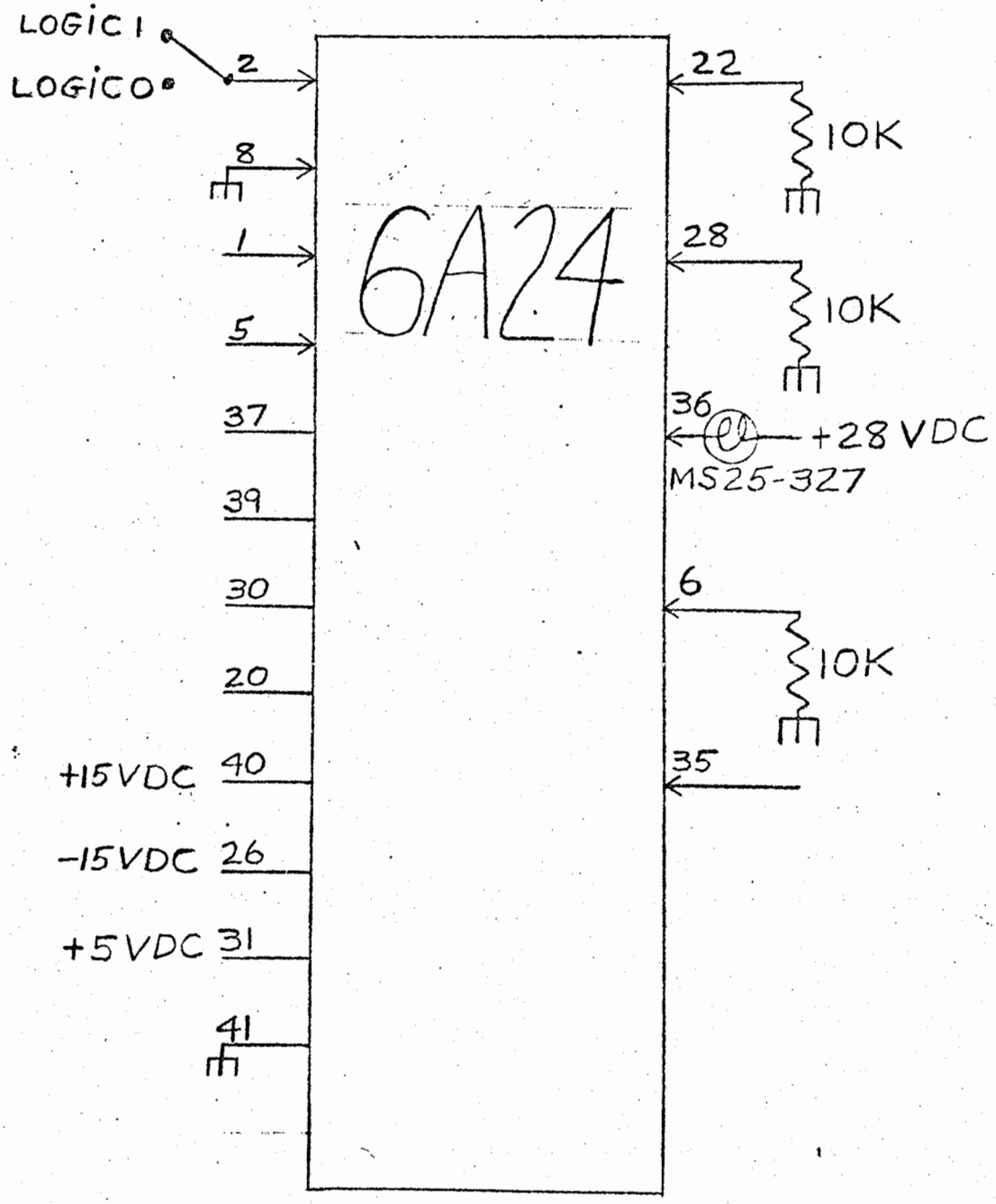


figure 1

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Remove Logic 1 from Pin 5. With the application of a Logic 1 to Pin 1, a 10 nanosecond Logic 1 pulse to Pin 5, and reset "0" at Pin 2, the output voltage on Pin 22 shall be 0V ±50mV. Apply another 10 nanosecond Logic 1 pulse to Pin 5, output Pin 22 voltage shall be 5mV ±1mV greater than the previous value. Apply several more 10 nanosecond Logic 1 pulses to Pin 5, reading the output voltage on Pin 22 after each pulse. Each successive Pin 22 reading shall be 5mV ±1mV greater than the previous reading.

Apply a 60Hz ± 1Hz clock pulse to Pin 5. Output Pin 22 voltage shall increase to +10V ±.25V, then increase towards 0V.

4.2.2.3.3 Azimuth Zero Network Test.- When the inputs are applied as stated the azimuth zero indicating lamp (Pin 36) and also output Pins 35 & 36 shall function as described below.

Azimuth Zero Network Test Procedure

With a logic 1 applied to Pin 1, a 15Hz ± 1Hz clock pulse to Pins 5, 39, & 37, reset "0" at Pin 2, the output voltage on Pin 22 shall increase towards +8.92V ± 20mV. When Pin 22 reaches +8.29V ± 20mV, the azimuth indicator lamp (Pin 36) shall start blinking and continue blinking and continue blinking until output Pin 22 decreases to -8.92V ±20mV at which time the lamp shall go out.

Reset "0" at Pin 2. Output Pin 22 shall increase, when Pin 22 reaches +8.92 ±20mV, output Pin 35 shall switch from Logic 1 to Logic 0 where it shall remain until Pin 22 decreases to -8.92V ±20mV at which time Pin 35 shall switch back to Logic 1.

Reset "0" at Pin 2. Output Pin 22 shall increase, when Pin 22 reaches +8.92V±20mV, output Pin 6 shall switch from +5V±.1V to 0V where it shall remain until Pin 22 decreases to -8.92V±20mV at which time Pin 6 shall switch back to +5V±.1V.

5. PREPARATION FOR DELIVERY

5.1 Packaging.- Packaging and marking shall be in accordance with the requirements of Packaging Data Sheet (PDS), AMC Form 1029, 11746286.

6. NOTES

6.1 The notes specified in the general specification MIL-C-48411 are applicable to this specification.

Custodian:

Army - MU

Preparing Activity:

Army - MU

Project No. 1240-A614

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